

LINUX ASSIGNMENT

Major Steps & Commands

Q1. Multi-Team Project Server

```
[root@jp-linux-assignment03 ~]# tree /opt/project/
/opt/project/
├── release
│   ├── builds
│   │   └── application-v1.tar
│   ├── documentation
│   │   └── release-notes.txt
│   └── source
│       ├── app
│       │   └── application.sh
│       ├── config
│       │   └── database.conf
│       └── testing
│           ├── logs
│           │   └── test.log
│           └── reports
│               └── test-report.txt
└── 9 directories, 6 files
[root@jp-linux-assignment03 ~]#
```

```
[root@jp-linux-assignment03 ~]# tail -4 /etc/passwd
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
kabir:x:1001:1001::/home/kabir:/bin/bash
meera:x:1002:1002::/home/meera:/bin/bash
aarav:x:1003:1003::/home/aarav:/bin/bash
[root@jp-linux-assignment03 ~]# tail -4 /etc/group
aarav:x:1003:
developers:x:1004:
testers:x:1005:
managers:x:1006:
[root@jp-linux-assignment03 ~]#
```

Q2. Permission Design

```
[root@jp-linux-assignment03 ~]# ls -ld /opt/project/*
drwxr-x---. 4 kabir managers 41 Aug 31 18:18 /opt/project/release
drwxr-x---. 4 aarav developers 31 Aug 31 18:18 /opt/project/source
drwxrwx---. 4 meera testers 33 Aug 31 18:18 /opt/project/testing
[root@jp-linux-assignment03 ~]#
```

Q3. File-Level Permissions

```
[root@jp-linux-assignment03 ~]# ls -ld /opt/project/source/app/application.sh
-rwxr-x---. 1 aarav developers 0 Aug 31 18:18 /opt/project/source/app/application.sh
[root@jp-linux-assignment03 ~]# ls -ld /opt/project/source/config/database.conf
-rw-----. 1 aarav developers 0 Aug 31 18:18 /opt/project/source/config/database.conf
[root@jp-linux-assignment03 ~]# ls -ld /opt/project/testing/reports/test-report.txt
-rw-rw----. 1 meera testers 0 Aug 31 18:18 /opt/project/testing/reports/test-report.txt
[root@jp-linux-assignment03 ~]# ls -ld /opt/project/testing/logs/test.log
-rw-r-----. 1 meera testers 0 Aug 31 18:18 /opt/project/testing/logs/test.log
[root@jp-linux-assignment03 ~]# ls -ld /opt/project/release/builds/application-v1.tar
-rw-r-----. 1 kabir managers 0 Aug 31 18:18 /opt/project/release/builds/application-v1.tar
[root@jp-linux-assignment03 ~]# ls -ld /opt/project/release/documentation/release-notes.txt
-rw-r--r--. 1 kabir managers 0 Aug 31 18:18 /opt/project/release/documentation/release-notes.txt
[root@jp-linux-assignment03 ~]#
```

Q4. ACL Challenge

```
[root@jp-linux-assignment03 ~]# getfacl /opt/project/source
getfacl: Removing leading '/' from absolute path names
# file: opt/project/source
# owner: aarav
# group: developers
user::rwx
user:riya:r-x
group::r-x
mask::r-x
other::---
```

```
[root@jp-linux-assignment03 ~]# getfacl /opt/project/testing
getfacl: Removing leading '/' from absolute path names
# file: opt/project/testing
# owner: meera
# group: testers
user::rwx
user:riya:r-x
group::rwx
mask::rwx
other::---
```

```
[root@jp-linux-assignment03 ~]# getfacl /opt/project/release
getfacl: Removing leading '/' from absolute path names
# file: opt/project/release
# owner: kabir
# group: managers
user::rwx
user:riya:r--
group::r-x
mask::r-x
other::---
```

Q5. ACL Mask Trap

```
[root@jp-linux-assignment03 ~]# sudo setfacl -m u:riya:rw /opt/project/source/config/database.conf
[root@jp-linux-assignment03 ~]# getfacl /opt/project/source/config/database.conf
getfacl: Removing leading '/' from absolute path names
# file: opt/project/source/config/database.conf
# owner: aarav
# group: developers
user::rw-
user:riya:rw-
group:---
mask::rw-
other:---

[root@jp-linux-assignment03 ~]# sudo setfacl -m m:rw /opt/project/source/config/database.conf
[root@jp-linux-assignment03 ~]# getfacl /opt/project/source/config/database.conf
getfacl: Removing leading '/' from absolute path names
# file: opt/project/source/config/database.conf
# owner: aarav
# group: developers
user::rw-
user:riya:rw-
group:---
mask::rw-
other:---
```

Q6. Conflicting Requirements

```
[root@jp-linux-assignment03 ~]# ls -ld /opt/project/testing/reports
drwxrwx---+ 2 meera testers 29 Aug 31 18:18 /opt/project/testing/reports
[root@jp-linux-assignment03 ~]# getfacl /opt/project/testing/reports
getfacl: Removing leading '/' from absolute path names
# file: opt/project/testing/reports
# owner: meera
# group: testers
user::rwx
user:riya:rwx
group::rwx
mask::rwx
other::---
```

Q7. Remove Specific ACL Entry

```
[root@jp-linux-assignment03 ~]# sudo setfacl -x u:riya /opt/project/testing/reports
[root@jp-linux-assignment03 ~]# getfacl /opt/project/testing/reports
[
getfacl: Removing leading '/' from absolute path names
# file: opt/project/testing/reports
# owner: meera
# group: testers
user::rwx
group::rwx
mask::rwx
other::---
```

Q8. Recursive ACL

```
[root@jp-linux-assignment03 ~]# getfacl /opt/company/finance
getfacl: Removing leading '/' from absolute path names
# file: opt/company/finance
# owner: root
# group: root
user::rwx
user:riya:r-x
group::r-x
mask::r-x
other::r-x

[root@jp-linux-assignment03 ~]# getfacl /opt/company/finance/salary
getfacl: Removing leading '/' from absolute path names
# file: opt/company/finance/salary
# owner: root
# group: root
user::rwx
user:riya:r-x
group::r-x
mask::r-x
other::r-x
```

```
[root@jp-linux-assignment03 ~]# getfacl /opt/company/finance/invoices
getfacl: Removing leading '/' from absolute path names
# file: opt/company/finance/invoices
# owner: root
# group: root
user::rwx
user:riya:r-x
group::r-x
mask::r-x
other::r-x
```

```
[root@jp-linux-assignment03 ~]# getfacl /opt/company/finance/reports
getfacl: Removing leading '/' from absolute path names
# file: opt/company/finance/reports
# owner: root
# group: root
user::rwx
user:riya:r-x
group::r-x
mask::r-x
other::r-x
```

Q9. Default ACL

```
[root@jp-linux-assignment03 ~]# getfacl /opt/company/common/test.txt
getfacl: Removing leading '/' from absolute path names
# file: opt/company/common/test.txt
# owner: root
# group: root
user::rw-
user:riya:rw-
group::r-x          #effective:r--
mask::rw-
other::r--
```

Q10. Diagnose the Problem

```
[root@jp-linux-assignment03 ~]# getfacl /opt/project/testing
getfacl: Removing leading '/' from absolute path names
# file: opt/project/testing
# owner: meera
# group: testers
user::rwx
user:riya:r-x
group::rwx
mask::rwx
other::---
```

Q11. Ownership + ACL Combination

```
[root@jp-linux-assignment03 ~]#  
[root@jp-linux-assignment03 ~]# ls /var/www/company/  
config private public  
[root@jp-linux-assignment03 ~]#
```

```
[root@jp-linux-assignment03 ~]# ls -ld /var/www/company/public  
drwxr-xr-x. 2 aarav developers 6 Aug 31 18:46 /var/www/company/public  
[root@jp-linux-assignment03 ~]# ls -ld /var/www/company/private  
drwxr-x---+ 2 kabir managers 6 Aug 31 18:46 /var/www/company/private  
[root@jp-linux-assignment03 ~]# ls -ld /var/www/company/config  
drw-rw----+ 2 aarav developers 6 Aug 31 18:46 /var/www/company/config  
[root@jp-linux-assignment03 ~]#
```

Q12. Find the Correct Command

```
chown meera report.txt  
chgrp testers report.txt  
chown aarav:developers report.txt  
chmod 750 report.txt  
setfacl -m u:riya:rx report.txt  
setfacl -x u:riya report.txt  
getfacl report.txt  
setfacl -m d:u:riya:rw directory
```

Q13. Troubleshooting Challenge

```
[root@jp-linux-assignment03 ~]# ls -ld /opt/project/release  
drwxr-x---+ 4 kabir managers 41 Aug 31 18:18 /opt/project/release  
[root@jp-linux-assignment03 ~]# getfacl /opt/project/release  
getfacl: Removing leading '/' from absolute path names  
# file: opt/project/release  
# owner: kabir  
# group: managers  
user::rwx  
user:riya:r--  
group::r-x  
mask::r-x  
other::---  
  
[root@jp-linux-assignment03 ~]#
```

Q14. Real Administrator Scenario

```
[root@jp-linux-assignment03 ~]# ls -l /opt/application  
total 0  
drwxrwx---+ 2 aarav developers 6 Aug 31 18:54 documentation  
drwxr-x---. 2 manager1 developers 6 Aug 31 18:54 production  
drwxr-x---. 2 aarav developers 6 Aug 31 18:54 source  
drwxr-x---. 2 riya testers 6 Aug 31 18:54 testing  
[root@jp-linux-assignment03 ~]#
```

```
[root@jp-linux-assignment03 ~]#  
[root@jp-linux-assignment03 ~]# getfacl /opt/application/source  
getfacl: Removing leading '/' from absolute path names  
# file: opt/application/source  
# owner: aarav  
# group: developers  
user::rwx  
group::r-x  
other::---
```

```
[root@jp-linux-assignment03 ~]#
[root@jp-linux-assignment03 ~]#
[root@jp-linux-assignment03 ~]# getfacl /opt/application/testing
getfacl: Removing leading '/' from absolute path names
# file: opt/application/testing
# owner: riya
# group: testers
user::rwx
group::r-x
other::---
```

```
[root@jp-linux-assignment03 ~]#
[root@jp-linux-assignment03 ~]#
[root@jp-linux-assignment03 ~]# getfacl /opt/application/documentation
getfacl: Removing leading '/' from absolute path names
# file: opt/application/documentation
# owner: aarav
# group: developers
user::rwx
user:manager1:rw-
group::r-x
mask::rwx
other::---
```

```
[root@jp-linux-assignment03 ~]#
[root@jp-linux-assignment03 ~]#
[root@jp-linux-assignment03 ~]# getfacl /opt/application/production
getfacl: Removing leading '/' from absolute path names
# file: opt/application/production
# owner: manager1
# group: developers
user::rwx
group::r-x
other::---
```

Q15. Final Boss

```
[root@jp-linux-assignment03 ~]# ls -ld /opt/company/development
drwxrwx---+ 4 root root 41 Aug 31 18:59 /opt/company/development
[root@jp-linux-assignment03 ~]# ^C
[root@jp-linux-assignment03 ~]# ls -ld /opt/company/development
drwxrwx---+ 4 root root 41 Aug 31 18:59 /opt/company/development
[root@jp-linux-assignment03 ~]# ls -ld /opt/company/deployment
drwxr-x---+ 3 manager1 management 22 Aug 31 18:59 /opt/company/deployment
[root@jp-linux-assignment03 ~]# ls -ld /opt/company/shared
drwxrwx---. 2 root root 6 Aug 31 18:59 /opt/company/shared
```

```
[root@jp-linux-assignment03 ~]# getfacl /opt/company/development
getfacl: Removing leading '/' from absolute path names
# file: opt/company/development
# owner: root
# group: root
user::rwx
user:manager1:r-x
group::rwx
mask::rwx
other::---
```

```
[root@jp-linux-assignment03 ~]# getfacl /opt/company/testing
getfacl: Removing leading '/' from absolute path names
# file: opt/company/testing
# owner: tester1
# group: testing
user::rwx
group::rwx
other::---
```

```
[root@jp-linux-assignment03 ~]#
```

```
[root@jp-linux-assignment03 ~]# getfacl /opt/company/deployment
getfacl: Removing leading '/' from absolute path names
# file: opt/company/deployment
# owner: manager1
# group: management
user::rwx
user:tester1:r-x
group::r-x
mask::r-x
other::---
```

```
[root@jp-linux-assignment03 ~]# getfacl /opt/company/deployment/releases
getfacl: Removing leading '/' from absolute path names
# file: opt/company/deployment/releases
# owner: manager1
# group: management
user::rwx
user:devops2:rw-
group::r-x
mask::rwx
other::---
```

```
[root@jp-linux-assignment03 ~]# getfacl /opt/company/shared
getfacl: Removing leading '/' from absolute path names
# file: opt/company/shared
# owner: root
# group: root
user::rwx
group::rwx
other::---
```